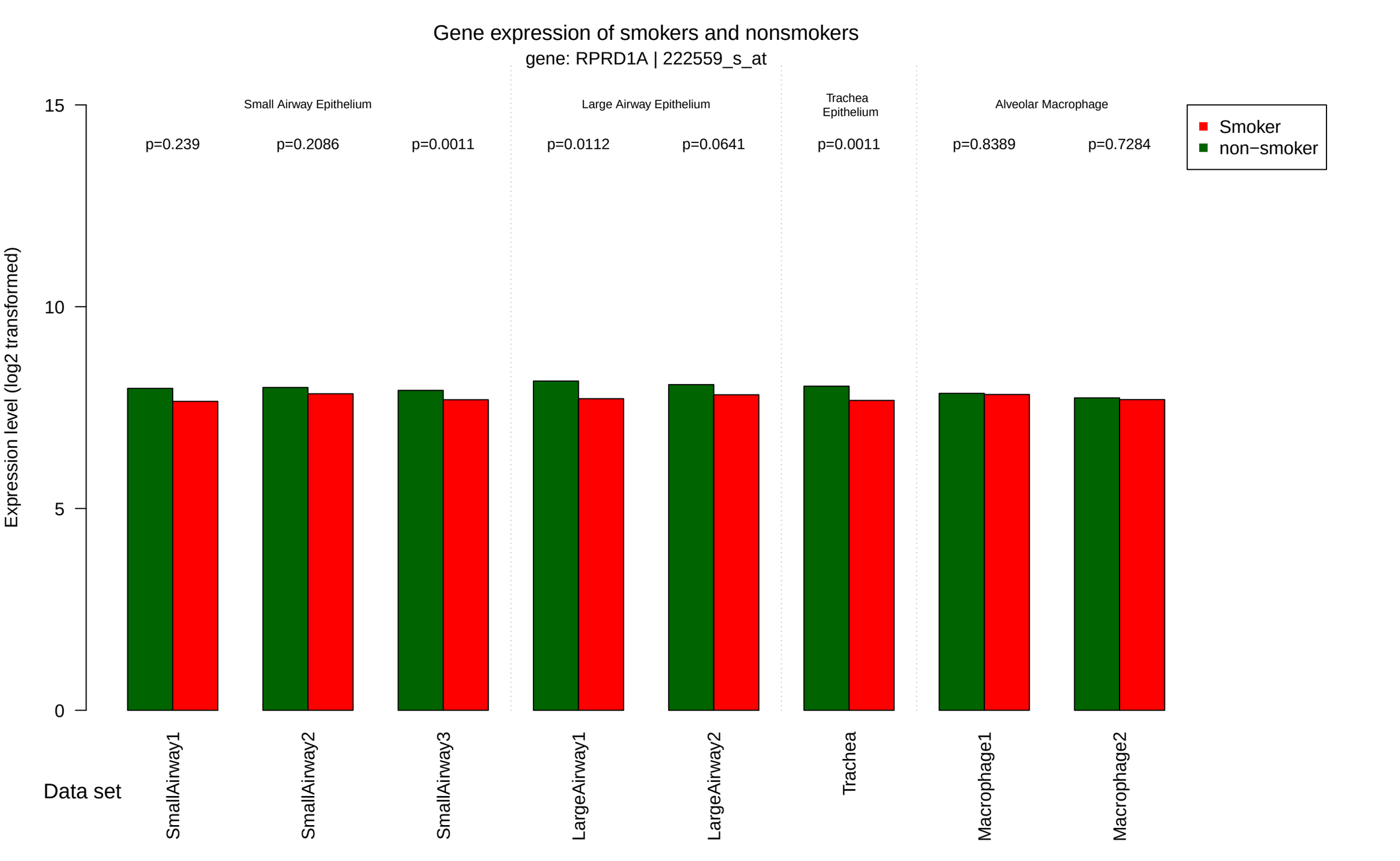
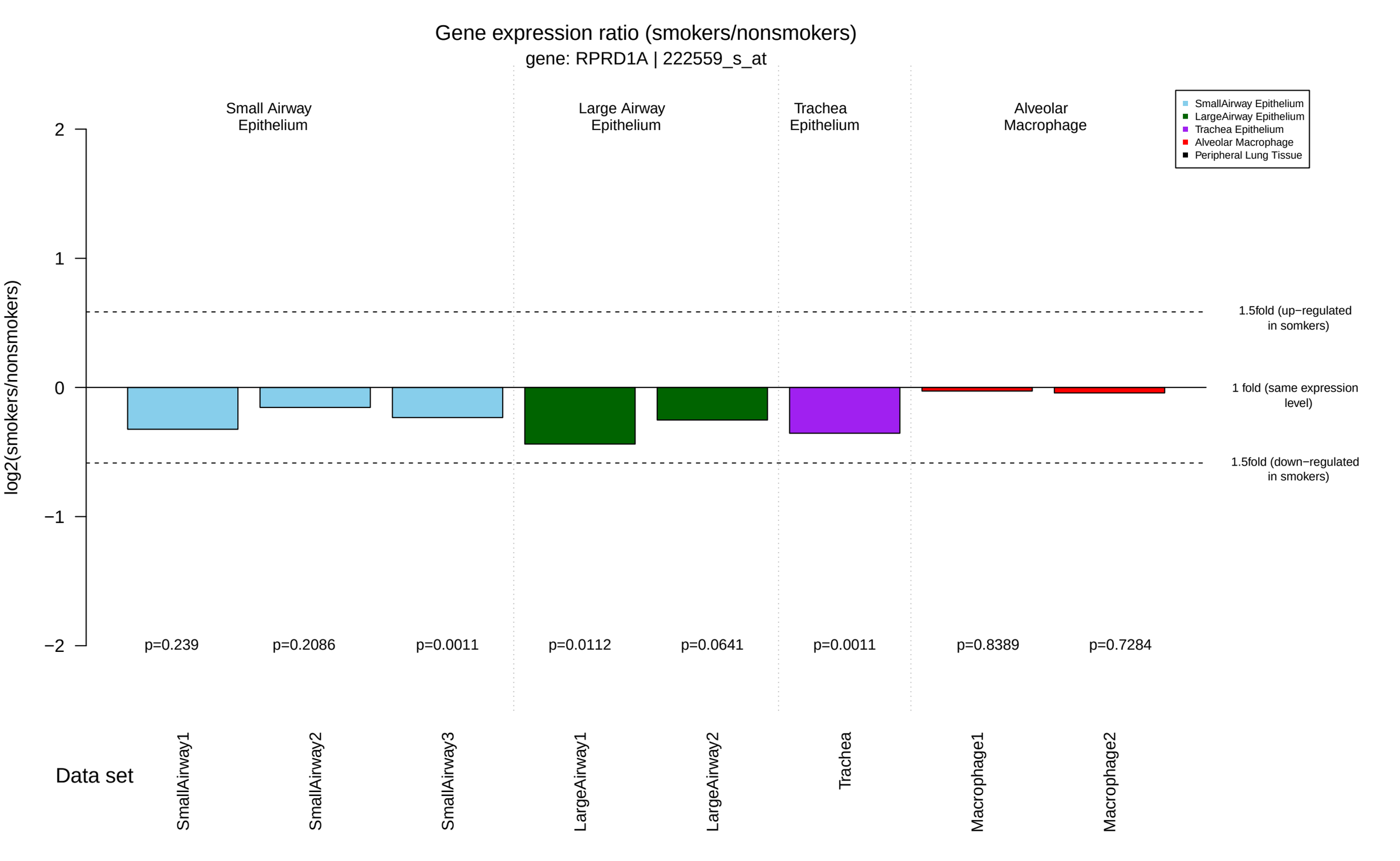
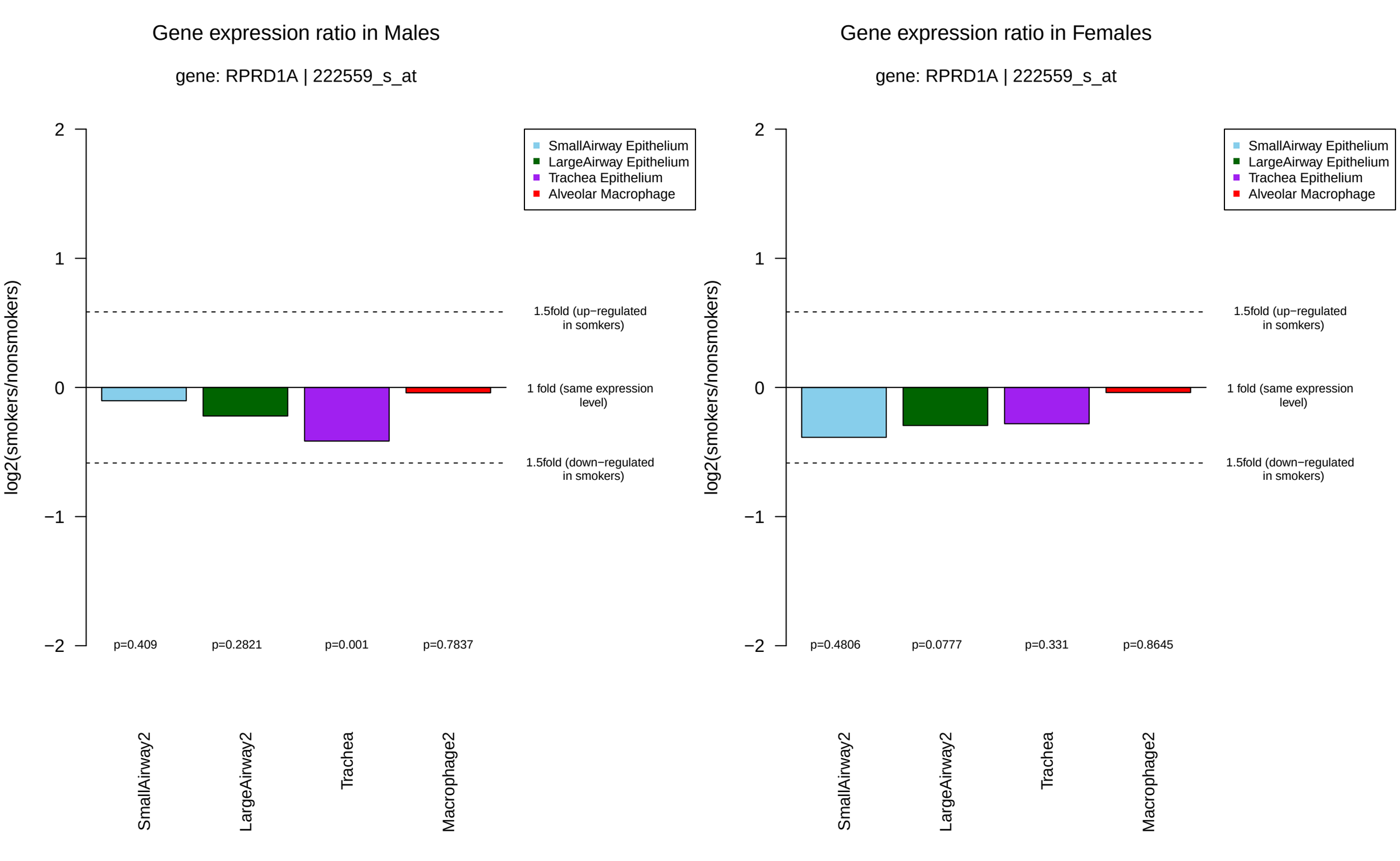
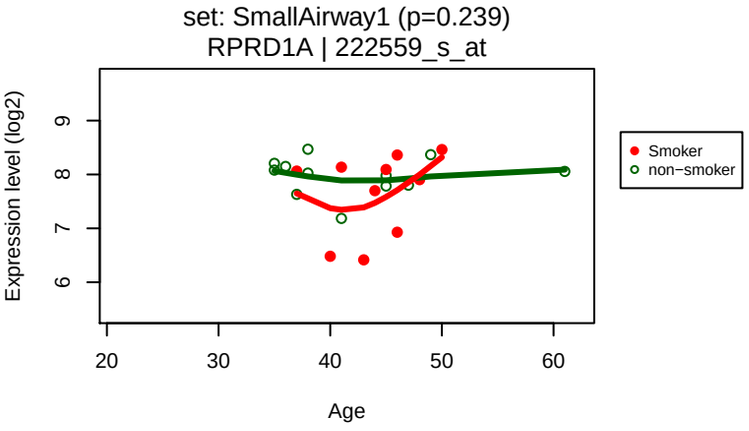
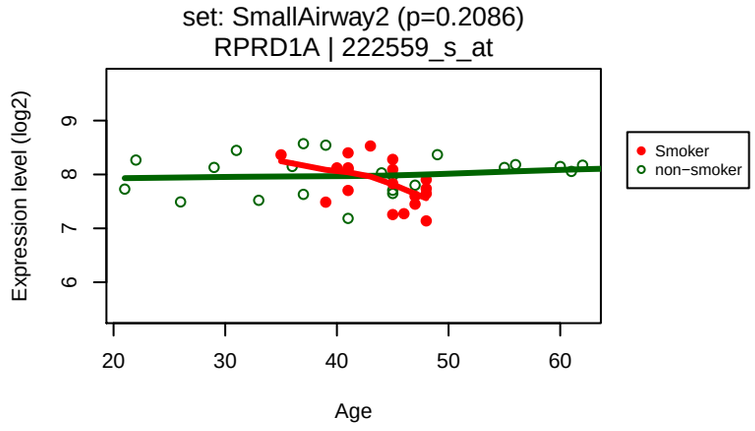
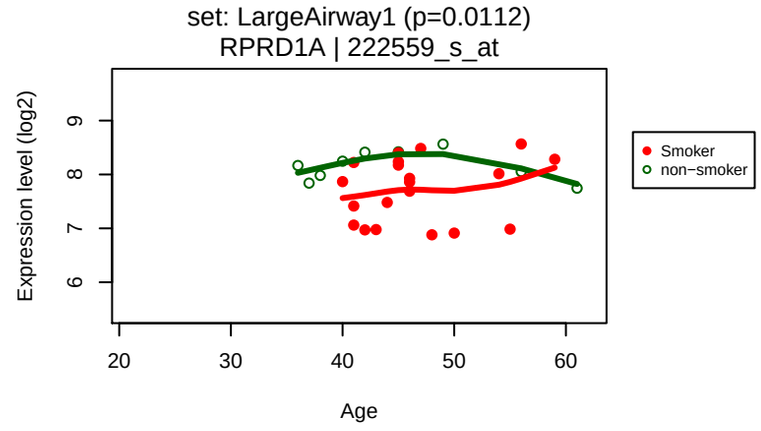

Gene expression of smokers and nonsmokers
gene: RPRD1A | 222559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.239
p=0.2086
p=0.0011
p=0.0112
p=0.0641
p=0.0011
p=0.8389
p=0.7284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

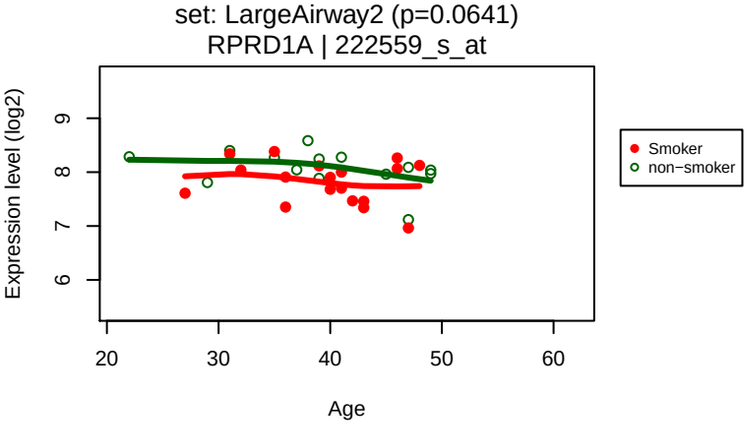

Gene expression ratio (smokers/nonsmokers)
gene: RPRD1A | 222559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.239
p=0.2086
p=0.0011
p=0.0112
p=0.0641
p=0.0011
p=0.8389
p=0.7284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

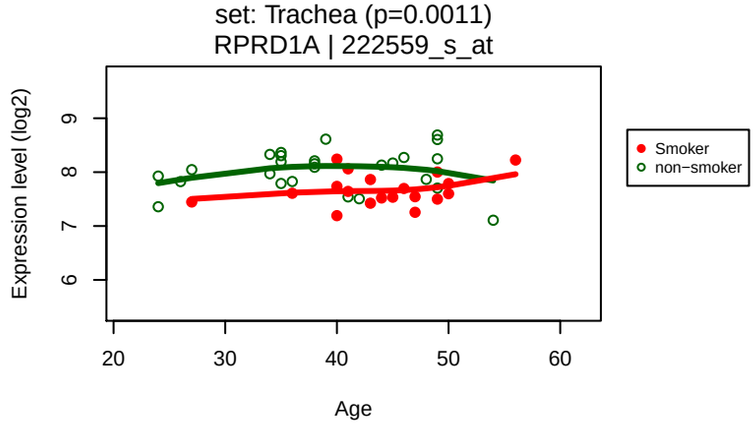

Gene expression ratio in Males
gene: RPRD1A | 222559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.409
p=0.2821
p=0.001
p=0.7837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD1A | 222559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4806
p=0.0777
p=0.331
p=0.8645
SmallAirway2
LargeAirway2
Trachea
Macrophage2

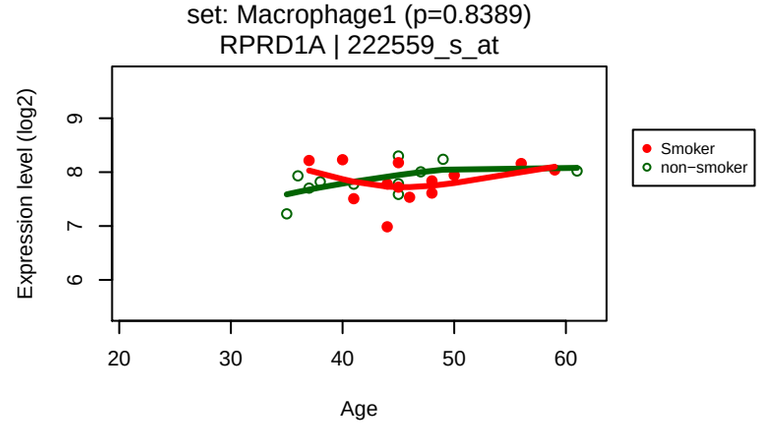

set: SmallAirway1 (p=0.239)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker

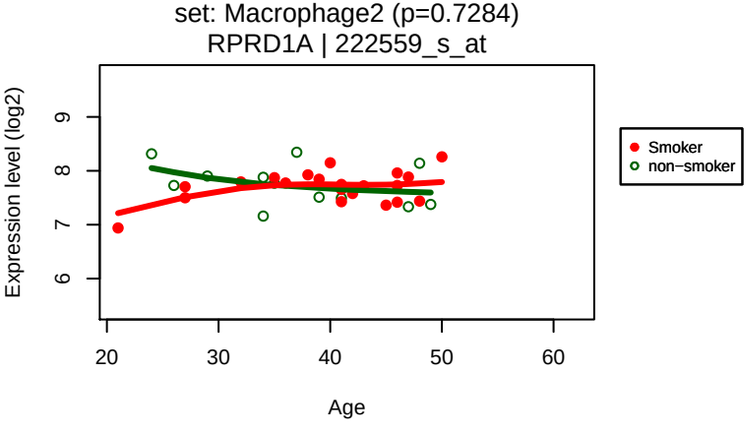

set: SmallAirway2 (p=0.2086)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker

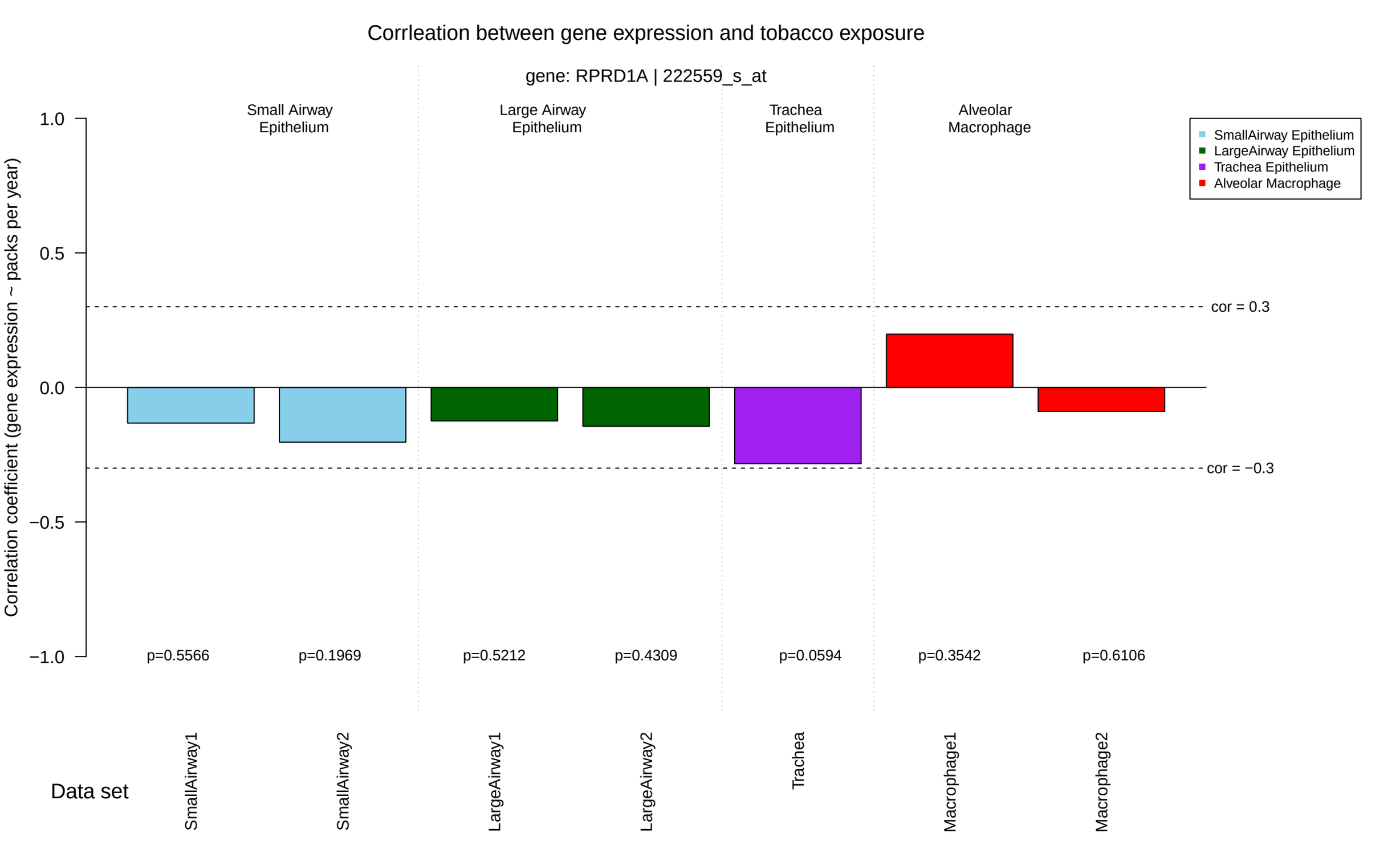

set: LargeAirway1 (p=0.0112)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0641)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8389)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7284)
RPRD1A | 222559_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD1A | 222559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5566
p=0.1969
p=0.5212
p=0.4309
p=0.0594
p=0.3542
p=0.6106
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2